Single-cell deep neural profiling identifies a tumor cell-specific gene target in head and neck squamous cell carcinoma: HPV-stratified validation in a Bangladeshi cohort

1. Background

Head and neck squamous cell carcinoma (HNSCC) is a heterogeneous malignancy accounting for over 90% of head and neck cancers, with human papillomavirus (HPV) infection representing a major etiological factor, particularly in oropharyngeal subsites. HPV-positive and HPV-negative HNSCC are increasingly recognized as biologically distinct entities with differing tumor microenvironments (TME), treatment responses, and prognoses. Despite this, five-year overall survival remains below 50%, and relapse rates exceed 50%, underscoring the need for improved molecular stratification.

Bulk transcriptomic approaches (e.g., TCGA) have identified molecular subtypes of HNSCC but cannot resolve cell-type-specific contributions within the tumor. Single-cell RNA sequencing (scRNA-seq) has enabled higher-resolution dissection of the TME. Foundational work by Puram et al. (2017) identified recurrent malignant cell expression programs in HNSCC, including a partial epithelial-mesenchymal transition (p-EMT) program associated with invasion and metastasis, demonstrating that clinically meaningful heterogeneity exists within the malignant compartment itself, not only across cell types.

A recent unified single-cell atlas (Conde-Lopez et al., 2026, iScience) integrated 78 HNSCC patients (274,911 cells) across four public datasets (HNSCatlas resource) and reported that HPV status shapes overall immune and stromal cell composition, with HPV+ tumors enriched in adaptive immune populations and HPV− tumors showing more stromal/innate immune populations. Notably, this atlas focused on cell-type composition and did not perform malignant-cell substate discovery at the gene level, and its findings did not remain significant after multiple-testing correction across the full cohort, leading the authors to explicitly call for validation in independent cohorts. Furthermore, the atlas contains no South Asian or Bangladeshi patient representation.

The candidate has prior experience applying deep generative (variational autoencoder-based) latent profiling to bulk transcriptomic data to uncover previously unrecognized regulator genes in cancer (e.g., RAP1GAP2 in oropharyngeal carcinoma). This pipeline extends that approach to the single-cell setting using scVI, applied specifically to the malignant (tumor) cell compartment of the HNSCatlas resource.

2. Rationale

- Existing HNSCC single-cell atlases characterize cell-type composition but do not resolve gene-level heterogeneity within the malignant compartment itself, leaving potential HPV-associated malignant cell substates unexplored.
- The reported HPV-associated immune/stromal shift is explicitly described by its original authors as hypothesis-generating and in need of independent validation, given the loss of significance after multiple-testing correction in the full cohort.
- No existing HNSCC single-cell or bulk resource includes Bangladeshi or South Asian patients; ethnic and geographic representation gaps limit generalizability of current findings.

- The candidate's own patient cohort includes pre-existing HPV genotyping data (from an ongoing HPV screening project), enabling HPV-stratified analysis without additional screening burden.
- A combined dry-lab (computational discovery and cross-cohort confirmation) and wet-lab (tissue-level qRT-PCR/IHC) design is feasible within the candidate's timeframe, computational resources, and laboratory access, while retaining translational value.

3. Objectives

General Objective

To identify malignant cell-state gene(s) in HNSCC through single-cell computational discovery and to validate their expression pattern, together with established HPV-associated immune/stromal markers, in a Bangladeshi patient cohort stratified by HPV status.

Specific Objectives

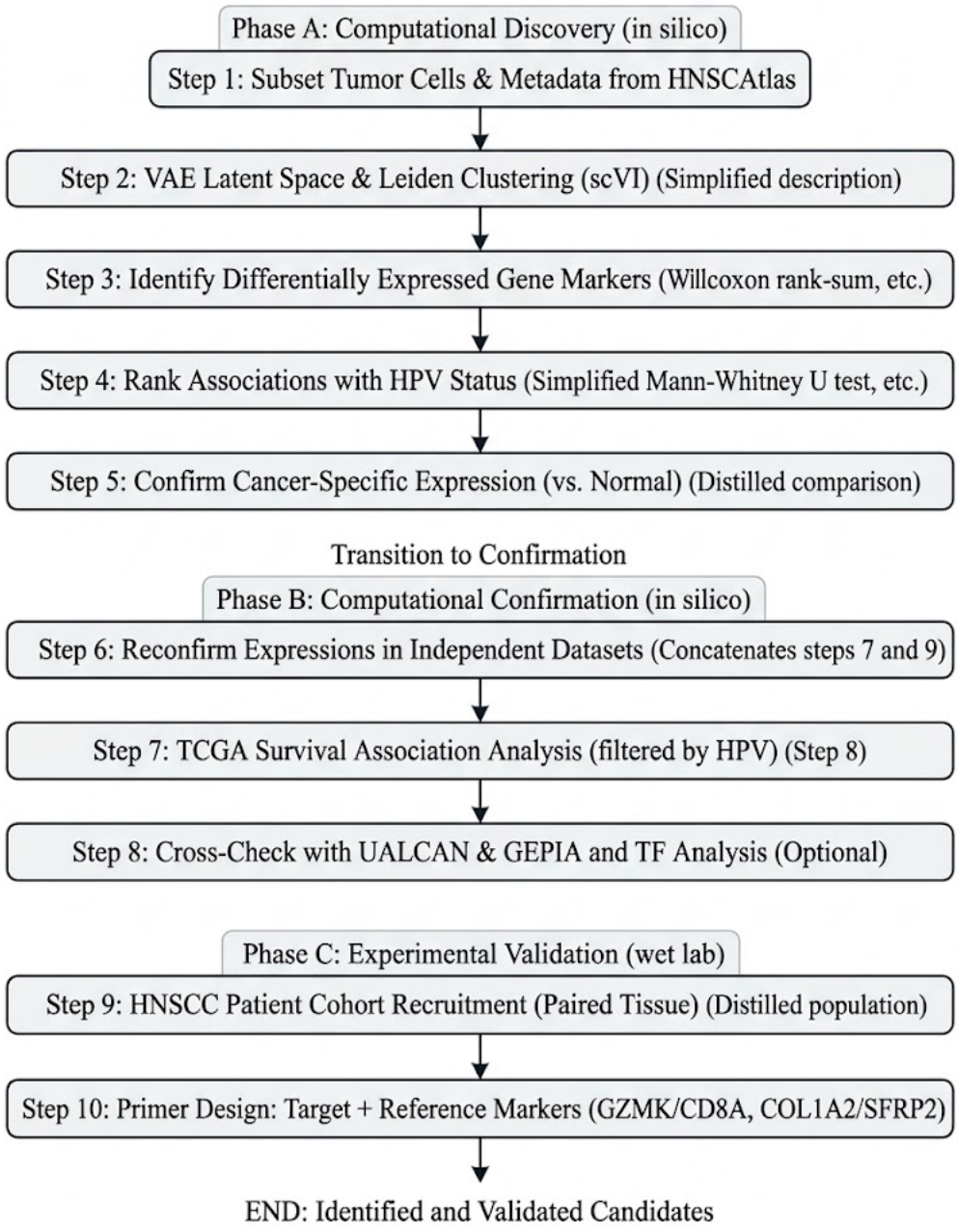
1. To computationally resolve malignant cell substates within the HNSCC tumor-cell compartment of the HNSCatlas resource using scVI-based latent representation learning and Leiden clustering.
2. To identify differentially expressed marker gene(s) defining these malignant substates and to assess their association with HPV status in silico.
3. To cross-validate candidate gene(s) using an independent public single-cell dataset and the TCGA-HNSC bulk cohort (expression and survival analysis via cBioPortal, UALCAN, and GEPIA).
4. To compare candidate gene expression between malignant/tumor cells and normal epithelial cells within the same atlas, confirming cancer-cell specificity prior to wet-lab validation.
5. To validate candidate gene expression by qRT-PCR in primary tumor versus adjacent normal tissue from a Bangladeshi HNSCC cohort, stratified by HPV status.
6. To assess established immune (e.g., GZMK, CD8A) and stromal/fibroblast (e.g., COL1A2, SFRP2) marker gene expression in the same cohort to test, for the first time in a South Asian population, whether the reported HPV-associated immune-versus-stromal TME shift is reproducible.
7. (Optional, resource-permitting) To confirm candidate gene and/or immune marker protein expression by immunohistochemistry (IHC).

4. Possible Outcomes

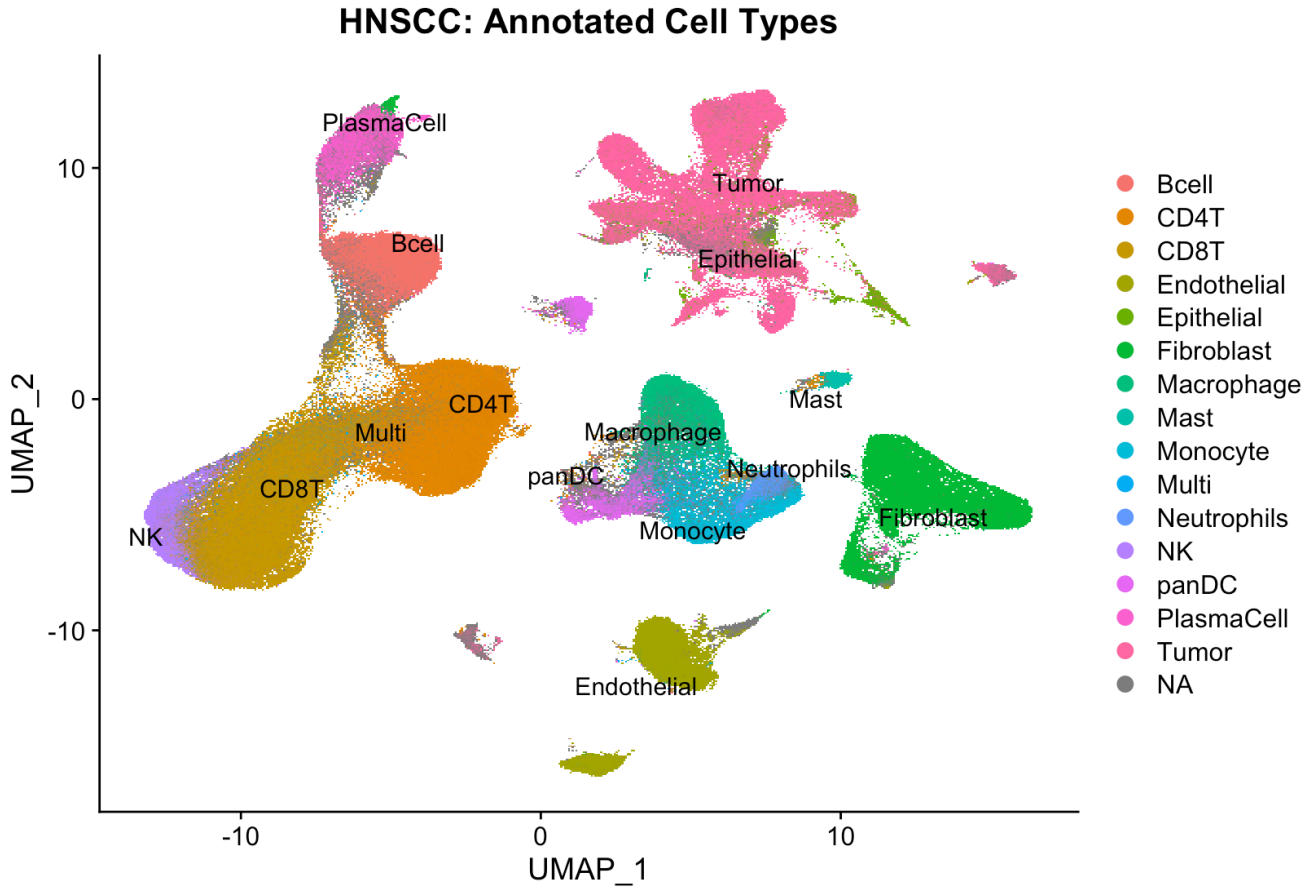
- Identification of a novel or previously underappreciated malignant cell-state gene in HNSCC, potentially associated with an invasive or HPV-linked cell state.
- Confirmation of the candidate gene's differential expression pattern in an independent Bangladeshi cohort, supporting its relevance beyond the discovery dataset.
- Either confirmation or refutation of the global HPV-associated immune/stromal TME shift in a South Asian population: both outcomes are scientifically valuable, given the hypothesis-generating status of the original finding.
- A candidate biomarker with potential prognostic relevance, supported by correlative TCGA survival data.
- Preliminary evidence, at the protein level (if IHC is performed), supporting mRNA-level findings.

- A publicly documented, reusable computational pipeline (scVI-based malignant substate discovery) applicable to future HNSCC or other squamous cell carcinoma studies.

5. Methodology



Test Run: single-cell RNA-seq of HNSCC



References

Bray, F., Ferlay, J., Soerjomataram, I., Siegel, R.L., Torre, L.A., and Jemal, A. (2018). Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J. Clin. 68, 394-424.

Conde-Lopez, C., Marripati, D., Besso, M.J., Roscher, M., Han, R., Hadiwikarta, W.W., Elkabets, M., Hess, J., and Kurth, I. (2026). A unified single-cell atlas of HNSCC: toward characterizing HPV- and sex-associated TME variability. iScience 29, 115863.

Lopez, R., Regier, J., Cole, M.B., Jordan, M.I., and Yosef, N. (2018). Deep generative modeling for single-cell transcriptomics. Nat. Methods 15, 1053-1058.

Puram, S.V., Tirosh, I., Parikh, A.S., Patel, A.P., Yizhak, K., Gillespie, S., Rodman, C., Luo, C.L., Mroz, E.A., Emerick, K.S., et al. (2017). Single-cell transcriptomic analysis of primary and metastatic tumor ecosystems in head and neck cancer. Cell 171, 1611-1624.e24.

Svensson, V., Gayoso, A., Yosef, N., and Pachter, L. (2020). Interpretable factor models of single-cell RNA-seq via variational autoencoders. Bioinformatics 36, 3418-3421.

Traag, V.A., Waltman, L., and van Eck, N.J. (2019). From Louvain to Leiden: guaranteeing well-connected communities. Sci. Rep. 9, 5233.

Way, G.P., and Greene, C.S. (2018). Extracting a biologically relevant latent space from cancer transcriptomes with variational autoencoders. Pac. Symp. Biocomput. 23, 80-91.